

MODULE-3

Interpretation and Report Writing

Meaning and technique of interpretation ;

Precautions in interpretation ;

Significance of report writing ;

Different steps in report writing ;

Layout of a research report ;

Types of report ;

Mechanics of writing a research report ;

Conclusion

Referencing in academic writing ;

Bibliography ;

Interpretation

- After the data is collected and analysed using several data analysis methods, the next task is to draw inferences from these data.
- In other words, Interpretation of data needs to be done, so as to derive certain conclusions, which is the whole purpose of the research study.
- Interpretation can be broadly understood as the process of converting collected data into meaningful conclusions by :
 - drawing inferences
 - identifying patterns
 - making connections between different aspects of information gathered during the research to generate meaningful results

Need for Interpretation ??

- Maintaining continuity of research
- Pointers for further research
- Communicate the significance of research
- Transition of exploratory research to experimental research
- It paves the way for the formulation of hypotheses for experimental research
- Through interpretation, the investigator can better understand why his /her findings are what they are
- It helps the researcher to make others to understand the real significance of his / her research findings
- It opens new avenues of intellectual venture and stimulates the hunt for more knowledge

Techniques of Interpretation

Steps

1. Explanation about the **relations** which the researcher has found during the research study
2. Interpretation of the **lines of relationship** in terms of the underlying processes
3. Finding **thread of uniformity** that lies beneath the diversified research findings and thereby, forming generalizations and theories.
4. **Extraneous information**, collected during the study, must also be considered while interpreting the final results of research study, as it may result in better understanding of the problem in hand.

5. **Consulting an expert** having an insight of the research study who points out omissions and errors in logical argumentation
6. This will result in correct interpretation and, add to **usefulness of the research** results.
7. Interpretation must be done bearing in mind **all relevant factors** affecting the problem to avoid false generalization.
8. All relevant factors affecting the problem must be considered before forming interpretations or generalisations. Otherwise it may lead to **incorrect conclusions.**

Precautions in Interpretation

1. Ensure proper data collection
2. Data Analysis
3. Errors can arise – be careful
4. Results of Hypothesis Testing
5. Statistical measures properly
6. Avoid broad generalizations
7. Quality of interpretation

Difference Between Analysis & Interpretation

- Analysis helps in describing the general trends in the data, and differences and similarities among data points
- Interpretation, on the other hand, relates data to the objectives of the research. It explores relationships between multiple measures, qualifies and amplifies data, draws inferences and evaluates
- Analysis is made on the basis of observed data to identify the pattern. This trend or pattern gives useful information about the design or system
- Interpretation involves things like predicting the outcome, observing the output response and associating meaning to that output. In other words, making the numerical data speak is the purpose of Interpretation

- **Analysis** is the information deciphered after collecting the data (Trend or routine)
- **Interpretation** explains the reasons behind the trend or a prevailing practice (Why the trend?)
- **For instance :-**
 - Decipher the patterns of monsoon over several years refers to Analysis
 - The reasons behind this trend will be explained through Interpretation

Research Report Writing

- Dr. Shlesinger & M. Stephenson in the encyclopaedia of Social Sciences define Research as :
 - manipulation of things, concepts or symbols
 - for the purpose of generalizing
 - to extend correct or verify knowledge aids
 - in construction of theory or in the practice of an art
- ***Writing the Research Report is the last step of Research Process.***

Definition:

- Research report is defined as "Systematic, articulate and orderly presentation of research work in a written form"
- RR is a "research document" that contains basic and important aspects of research

Contents of Research Report

- Description of events
- Facts discovered during investigation
- Description of procedures/ analysis tools
- References carefully documented
- Conclusion
- Results objectively recorded
- Recommendations/ Suggestions

Objectives of Research Report

- To become familiar with the given subject and achieve new insights into it
- To portray the exact features or nature of an individual, group or object
- To determine the frequency with which something occurs or is associated with something else
- To study and test the relationship between two or more variables

Characteristics of an ideal Research Report

- Proper title
- Relevant subheadings and paragraph divisions
- The research report should be factual
- The research report should relate to certain period
- The research report should be clear, brief and concise
- The reporting must be prompt
- Distinguish between controllable and non-controllable factors

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- Appropriate remarks
 - Periodically reviewed
 - Permissible degree of inaccuracy
 - Highlight important deviations from standards
 - Visual reporting with pictures and diagrams
 - Comparison between comparable matters

Significance of Report Writing

- Major component of research
- Findings are brought into limelight
- Medium to communicate research work with relevant people
- Contributes to the body of knowledge
- Effective way of conveying the research work
- Reference material
- Aid for decision making

Steps in Report Writing

- Analysis of the subject-matter
- Making of the final outline
- Making rough draft
- Polishing the rough draft
- Making of bibliography
- Writing the final draft

Format / Layout of Research Report

A. Preliminary Pages

B. Main Text

1. Introduction

2. Statement of findings and recommendation

3. Results

4. The implications drawn from the results

5. Summary

C. End matter

Contents of the Research Report

- Cover page
- Summary
- Acknowledgement
- Index
- Introduction
- Objectives
- Literature review
- Research Methodology
- Data analysis and interpretation
- Conclusion
- Bibliography
- Appendix/Annexure

Precautions in Writing the Research Report

- Communicate research findings properly
- Length of the research report
- Curiosity within reader
- Objective style of writing
- Use of charts, graphs and tables
- Appropriate layout
- Avoid grammatical mistakes
- Logical analysis
- Appendices
- Bibliography of necessary sources
- Appearance
- Policy implications
- Originality
- Objectives of study

Research Paper Writing

Definition

- According to Berge and Saffioti, "Research paper is a paper written to reflect a search that will present information to support a point of view on a particular topic".
- A research paper entails surveying a field of knowledge so as to find the best possible information in that field

Structure of a Research Paper

1. Abstract
2. Introduction
3. Review of Literature
4. Research methods and procedures
5. Analysis and findings
6. Conclusions and recommendations
7. Limitations and future scope
8. References

Referencing Styles

- The different references used by the researcher for the purpose of research study such as books, articles, other research papers, Internet URL's should be cited in standard accepted formats.
- There are different referencing styles such as :-
 - APA (American Psychological Association)
 - MLA (Modern Language Association)
 - Harvard referencing style
 - Chicago style of referencing

Referencing Styles - Elements

- Name of author
- Title of article/research paper
- Name of journal /periodical
- Volume number
- Date/month of issuance
- Year of publication
- Page number
- Name of book (in case of a book)
- Book publisher and place (in case of a book)

Referencing Styles

1. APA (American Psychological Association) Referencing Style:

The style to reference is :-

For a Research paper/Article

Authors surname followed by his (their) initials (Year of publication) Article title,
Name of Journal, Volume, Page no.

Example

Tomaszewski, S. & Showerman, S. (2010). IFRS in United States: Challenges and opportunities, Review of Business, 30, 59-71

2. MLA (mainly for a book):

Author's surname, Initials (Year of publication) Book title, Place: Publisher

Example

Kumar, R (2014), Research Methodology: A Step-by-Step Guide for Beginners(4th ed).

Thousand Oaks, California: Sage Publications

3. Harvard Referencing Style:

The broad style to reference is

Author Surname, Initials. (Publication Newspaper/Magazine Name, Day, Month, Year) , Published, Page(s). 'Article title Available at: URL or DOI (Accessed date)

4. Chicago style of referencing:

Author's first and last name, book title, place of publication, publisher, year and page numbers

Albert Einstein, *The Meaning of Relativity* (Princeton: Princeton University Press, 1923), 44-45

Ethics in Report/Research Paper Writing

- Avoid Plagiarism
- Honesty
- Objectivity
- Integrity
- Alertness
- Openness
- Confidentiality
- Publication
- Social Responsibility

Bibliography

- A systematic list of works, such as books and journal articles, that are related to a specific topic
- It can also be referred to as a list of works cited
- Bibliographies can be found at the end of books and articles or research paper
- They can be organized by author, subject or date of publication
- The bibliography structure always includes citing the author's name, the title of the work, the year of publication and the publisher for each source, one consults for a paper or project
- Trillin, Calvin. "Culture Shopping." New Yorker 15 Feb. 1993: 48 – 51